NABIL AL-SHANTEER

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EXPERIENCE — *[Website]*

AI City Innovation Sandpit Eastleigh, UK: *Mar 2026 — Jun 2026*

AI Solutions Intern - [Project Article]

- Responsibilities include the discovery phase for both **Citizens Advice Eastleigh** and the British **National Health Service (NHS)** to learn their pain points and agree on user requirements.
- For **Citizens Advice**: Developed an AI Chatbot with self-updating RAG system to contain only the most up-to-date legal information regarding housing, benefits, and other sectors of law using the United Kingdom government's official GOV.UK API. Tailored responses in a legal format with source documentation, and connected front-end chatbot interface with back-end database management.
- For the **NHS**: Developed a survey feedback analysis tool that synthesises thousands of medical survey feedback. The pipeline anonymises responses, clusters them based on themes, and features a "human-in-the-loop" mode and "automatic" mode. Lastly, clusters are formatted in an interactive whiteboard for analysts to view, edit, and extract useful reports from.

Bloctopus London, UK: *Oct 2025 — Jan 2026*

Technical Lead, Orchestration & Routing Logic - [Project Article]

- Planned, designed, evaluated, & deployed **BloctoGPT**, an *AI-powered* conversational agent that allows non-expert users to design, deploy, & manage Web3 infrastructure on the **Bloctopus** platform, optimising AI agents' context windows - *[Context Bloating Demo]*.
- Applied *statistical tools* like the Jaccard Index & Co-Occurrence Matrices to rank and select high-utility, relevant Model Context Protocol (MCP) tools from a catalogue of 1,000+ tools.
- Engineered a scalable, cloud-deployed, multi-component processing framework integrating data pipelines including Google's Retrieval-Augmented Generation (RAG) tool, Anthropic's Large Language Models (LLM), & various *APIs*.
- Collaborated in a team via agile development using *GitLab* & *GitHub* for version control & sprint planning. Organised sprints around three design iterations (BM25 version, Agent-to-Agent version, Decomposer-RAG version).
- Communicated progress & experimental results with stakeholders on a weekly basis, translating technical findings into documentation. *[Project Poster]*

Academic Theses University of Southampton, UK

Masters Thesis: ***Event-Driven and Many-Threaded Architectures for High-Frequency Trading*** *Jan 2026 — May 2026*

- Surveyed 19 papers (2008–2025) on *FPGA acceleration*, *event-driven computing*, *neuromorphic chips*, & *spiking neural networks (SNNs)* for HFT, classifying each by methodology and evidence strength. *[Evaluation Report]*
- Critiqued SNN trading backtests (Sharpe ratios above 20 from 23 trading days, no transaction costs) and set out five open research problems, including cross-substrate power benchmarking and *agent-based market simulation* with hardware-heterogeneous traders.

Bachelors Thesis: ***Deep Learning for Activity Recognition in Surveillance*** *Sep 2024 — May 2025*

- Reproduced & optimised vision-language model **ActionCLIP** for *few-shot learning* under limited data and compute. *Fine-tuned* transformer-based architectures on a custom dataset, reaching **97.56% accuracy** & **ROC-AUC of 0.9913** - *[Evaluation Report]*.
- Cut false positives using *unsupervised learning*, *statistical regularisation* & *Mahalanobis distance* for *outlier detection*, with applications in *fraud detection*, *risk monitoring*, & *market anomaly identification*.

Asher Amman, Jordan: *Jul 2024 — Sep 2024*

Machine Learning Researcher (Intern)

- Collected & curated hundreds of gesture images to train a convolutional neural network converting real-time Arabic Sign Language into written Arabic text. Model evaluation followed.

EDUCATION

University of Southampton, Southampton, UK 2022 —2026

Master of Electronic Engineering with Artificial Intelligence. See *[Academic Transcript]* for Modules. First Class Honours

Mashrek International School, Amman, Jordan 2019 — 2021

IB Higher Level Mathematics, Physics, Computer Science: 6/7, 6/7, 7/7 Final Grade: 41/45

SKILLS

- **CLOUD DEVELOPMENT**: Golang, Docker, Google Kubernetes & Compute Engines, APIs // **MACHINE LEARNING**: Python, PyTorch, OpenCV, Pandas, Numpy, MatPlotLib, SciKit, HuggingFace, AI Agency & Automation // **LABORATORIES**: PCB Design and Test, Mathematical Modeling of Non-Linear Systems, Analytical Computation of State Estimates (RLS, LS), and more.
- Cross-functional finance and engineering skills, paired with strong presentation skills and resilience towards unfamiliar problems. Solution-driven & optimiser, efficient worker, maintainer of effective communication.
- Able to perform under high pressure. Adaptive to company changes, ability to multitask and prioritise, multiple displays of effective project management.
- Interested in the intersection of quantitative modeling, machine learning, and financial markets particularly in algorithmic trading, risk analytics, and data-driven strategy design. 3-year member of the *[Quantitative Trading Society]* at the University of Southampton.
- Coming from a Middle Eastern background with strong ties to the region, native speaker of Arabic & French, experience studying under British systems and adapting to said work ethic.